

11. 2.

易知下两题被积函数均二元连续

$$1. (2) \lim_{k \rightarrow 1.0} \int_0^{\frac{\pi}{2}} \sqrt{1-k^2 \sin^2 \varphi} d\varphi = \int_0^{\frac{\pi}{2}} \cos \varphi d\varphi = 1$$

$$(5) \lim_{y \rightarrow 0} \int_0^1 \frac{e^x \sin xy}{y+1} dy = 0$$

$$2. (2) \frac{\partial f}{\partial y} = \sqrt{1-x^2} \cdot e^{y\sqrt{1-x^2}}, \text{ 在定义域内连续}$$

$$g'(y) = \int_{\sin y}^{\cos y} \sqrt{1-x^2} \cdot e^{y\sqrt{1-x^2}} dx - \sin y \cdot e^{y\sqrt{1-x^2}} - \cos y \cdot e^{y\sqrt{1-x^2}}$$

$$3. (2) g'(a) = \int_0^{\frac{\pi}{2}} \frac{1}{\cos x} \left(\cos x \cdot \frac{1}{1+a \cos x} + \cos x \cdot \frac{1}{1-a \cos x} \right) dx$$

$$= \int_0^{\frac{\pi}{2}} \left(\frac{1}{1+a \cos x} + \frac{1}{1-a \cos x} \right) dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{2}{1-a^2 \cos^2 x} dx$$

$$= 2 \cdot \int_0^{\frac{\pi}{2}} \frac{\sec^2 x}{\sec^2 x - a^2} dx = 2 \cdot \int_0^{+\infty} \frac{dt}{t^2 + 1 - a^2} dt = 2 \cdot \frac{1}{\sqrt{1-a^2}} \arctan \frac{t}{\sqrt{1-a^2}} \Big|_0^{+\infty} = \frac{\pi}{\sqrt{1-a^2}}$$

$$\therefore g(a) = 2 \cdot \arcsin a$$

11.3

$$1. (2) |e^{-t^2 x^2} \cos x| \leq e^{-t^2 x^2}.$$

而 $\int_0^{+\infty} e^{-t^2 x^2} = -1$, 收敛, 故原积分一致收敛

$$(3) (b) |e^{-\alpha x} \sin x| \leq e^{-\alpha x}, \quad \int_0^{+\infty} e^{-\alpha x} = -1 \quad \text{一致收敛}$$

(ii) 不论 N 多大, 取 $A = 2k\pi > N$, $\alpha = \frac{1}{A}$,

$$\left| \int_A^{+\infty} e^{-\alpha x} \sin x \, dx \right| = \left| \int_A^{+\infty} e^{-\frac{x}{A}} \sin x \, dx \right| = \left| \int_1^{+\infty} A e^{-t} \sin At \, dt \right|$$

$$\begin{aligned} &= \left| \int_1^{+\infty} e^{-t} d(\cos At) \right| = \left| e^{-t} \cos At \Big|_1^{+\infty} + \int_1^{+\infty} (-\sin At) e^{-t} \, dt \right| \\ &= \left| e^{-1} + \frac{1}{A} \int_1^{+\infty} e^{-t} d(\sin At) \right| \\ &= \left| \frac{1}{e} + \frac{1}{A} \left(e^{-t} \sin At \Big|_1^{+\infty} + \int_1^{+\infty} (\sin At) e^{-t} \, dt \right) \right| \\ &= \left| \frac{1}{e} + \frac{1}{A} \int_1^{+\infty} e^{-t} \sin At \, dt \right| \end{aligned}$$

则令原式 $= \varphi(t)$, 则

$$|\varphi(t)| \geq \frac{1}{e} + \frac{1}{A^2} |\varphi(t)| > \frac{1}{e}$$

故发散

$$2. (1) \quad \frac{e^{-ax} - e^{-bx}}{x} = \int_b^a -e^{-xy} dy = \int_a^b e^{-xy} dy$$

$$\therefore \text{原式} = \int_0^{+\infty} dx \int_a^b e^{-xy} dy$$

$$\text{对 } f(y) = \int_0^{+\infty} e^{-xy} dx, \quad a \leq y \leq b, \quad e^{-xy} \leq e^{-ax}, \quad \text{而 } \int_0^{+\infty} e^{-ax} dx \text{ 收敛}$$

则一致收敛, 可换序

$$\text{原式} = \int_a^b dy \int_0^{+\infty} e^{-xy} dx$$

$$= \int_a^b \left(-\frac{1}{y} e^{-xy} \Big|_0^{+\infty} \right) dy = \int_a^b \frac{1}{y} dy = \ln \frac{b}{a}$$

$$(3) \text{ 令 } t = x + \frac{1}{4},$$

$$2x^2 + x + 1 = 2 \cdot \left(t - \frac{1}{4}\right)^2 + t - \frac{1}{4} + 1 = 2t^2 - t + \frac{1}{8} + t - \frac{1}{4} + 1 = 2t^2 + \frac{7}{8}$$

$$\therefore \text{原式} = \int_0^{+\infty} e^{-(2t^2 + \frac{7}{8})} dt$$

$$= e^{-\frac{7}{8}} \cdot \int_0^{+\infty} e^{-2t^2} dt$$

$$= e^{-\frac{7}{8}} \cdot \frac{1}{\sqrt{2}} \cdot \int_0^{+\infty} e^{-\left(\frac{\sqrt{2}}{2}t\right)^2} d\left(\frac{\sqrt{2}}{2}t\right)$$

$$= e^{-\frac{7}{8}} \cdot \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{2}}{2}$$

$$= \frac{\sqrt{2}}{4} e^{-\frac{7}{8}} \sqrt{2}$$

3. $f(t) = \int_0^{+\infty} e^{-x} \cdot \frac{\sin tx}{x} dx$, 易知 $\forall t \in \mathbb{R}, |e^{-x} \cdot \frac{\sin tx}{x}| \leq \frac{e^{-x}}{x}$, 而 $\int_0^{+\infty} e^{-x} dx$ 收敛,
 $\frac{1}{x} \downarrow$ 且 $\rightarrow 0$
 故 $\int_0^{+\infty} \frac{e^{-x}}{x} dx$ 收敛

故点点收敛.

$$\frac{\partial (e^{-x} \frac{\sin tx}{x})}{\partial t} = e^{-x} \cdot \cos tx, \quad |e^{-x} \cos tx| \leq e^{-x}, \text{ 而 } \int_0^{+\infty} e^{-x} dx \text{ 收敛}$$

故一致收敛

又有在 $t \in \mathbb{R}$ 上连续; 故可在积分号下求导

$$\begin{aligned} f'(t) &= \int_0^{+\infty} e^{-x} \cdot \cos tx \cdot dx \\ &= - \int_0^{+\infty} \cos tx \, de^{-x} \\ &= - e^{-x} \cdot \cos tx \Big|_0^{+\infty} + \int_0^{+\infty} e^{-x} \cdot d \cos tx \\ &= 1 - t \cdot \int_0^{+\infty} e^{-x} \cdot \sin tx \cdot dx \end{aligned}$$

$$\begin{aligned} \text{又有 } f'(t) &= \int_0^{+\infty} e^{-x} \cos tx \, dx = \frac{1}{t} \int_0^{+\infty} e^{-x} \cdot d \sin tx \\ &= \frac{1}{t} \cdot [e^{-x} \sin tx]_0^{+\infty} + \int_0^{+\infty} \sin tx \cdot e^{-x} \cdot dx \\ &= \frac{1}{t} \cdot \int_0^{+\infty} \sin tx \cdot e^{-x} \, dx \end{aligned}$$

$$\begin{aligned} \therefore & t \cdot \int_0^{+\infty} e^{-x} \sin tx \, dx + \int_0^{+\infty} e^{-x} \cos tx \, dx \\ &= (t^2+1) \cdot \int_0^{+\infty} e^{-x} \cos tx \, dx = 1 \end{aligned}$$

$$\therefore f'(t) = \frac{1}{t^2+1}$$

$$f(t) = \arctan t + C$$

$$\text{而 } f(0)=0, \text{ 则 } f(t) = \arctan t$$